

Digital Design SET 111

Lecture (04)-Number Systems, Operations, and Codes

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- ☐ **Binary Arithmetic**
- ☐ **Complements of Binary Numbers**
- ☐ **Hexadecimal Numbers**
- ☐ **Octal Numbers**
- ☐ **Conversion Map**



- ☐ **Binary Arithmetic**
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Binary Arithmetic (1/2)



❑ Binary arithmetic is essential in all **digital computers** and in many other types of digital systems

❑ Binary Addition (+)

❑ The **four** basic rules for adding binary digits (bits)

❑ Ex:- Add the binary numbers $(11)_2$ to $(01)_2$

❑ Ex:- Add the binary numbers $(111)_2$ to $(11)_2$

A	+	B	Sum	Carry out
0		0	0	0
0		1	1	0
1		0	1	0
1		1	0	1

Carry 1 1 1

A	1	1	1	
B			1	1
			+	
Result	1	0	1	0

Carry 1 1

A		1	1
B		0	1
		+	
Result	1	0	0

Carry in	+	A	+	B	Sum	Carry out
1		0		0	1	0
1		0		1	0	1
1		1		0	0	1
1		1		1	1	1

Binary Arithmetic (2/2)



❑ Binary arithmetic is essential in all digital computers and in many other types of digital systems

❑ Binary Subtraction (-)

❑ The **four** basic rules for subtracting binary digits (bits)

	A	-	B	Subtract
	0		0	0
1	0		1	1
	1		0	1
	1		1	0

❑ Special Case: ?

❑ A **borrow** is required in binary only when you try to subtract a 1 from a 0

❑ Ex:- Subtract $(011)_2$ from $(101)_2$

Right column:

$$1-1=0$$

Middle Column:

Borrow 1 from next column to the left, making a 10 in this column, then $10 - 1 = 1$

Left column:

When a 1 is borrowed a 0 is left, so $0 - 0 = 0$

$$\begin{array}{r}
 0 \\
 A 1 1 \\
 B 0 1 1 \\
 \hline
 0 1 0
 \end{array}$$

$$\begin{array}{r}
 0 \\
 A 1 0 \\
 B 0 1 \\
 \hline
 1
 \end{array}$$



- ☐ Binary Arithmetic
- ☐ Complements of Binary Numbers
- ☐ Hexadecimal Numbers
- ☐ Octal Numbers
- ☐ Conversion Map

Complements of Binary Numbers (1/5)



❑ The **1's complement** and the **2's complement** of a binary number are important because they permit the representation of **negative** numbers.

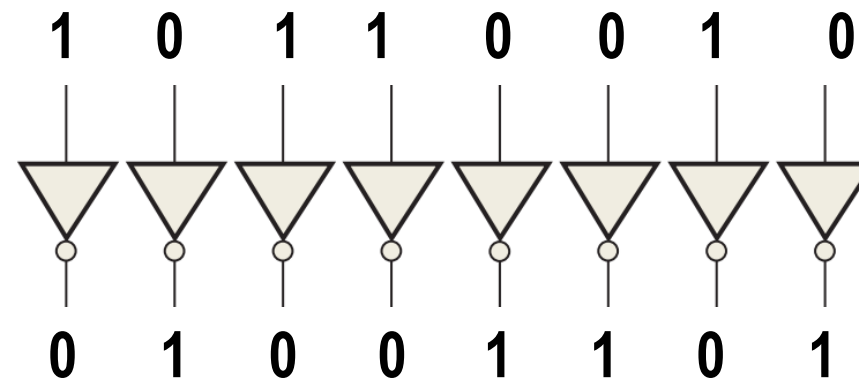
❑ 1's complement

❑ is obtained by changing each **1** $\rightarrow$ **0** and each **0** to **1**.

❑ Ex:- Find the 1's complement of $(10110010)_2$.

❑ In **hardware**, array of inverters can be used to provide the **1's complement**

Binary Number	1	0	1	1	0	0	1	0
	↓	↓	↓	↓	↓	↓	↓	↓
1's complement	0	1	0	0	1	1	0	1



Hardware Digital circuit for 1's complement

Complements of Binary Numbers (2/5)



□ 2's complement

□ 1st method:

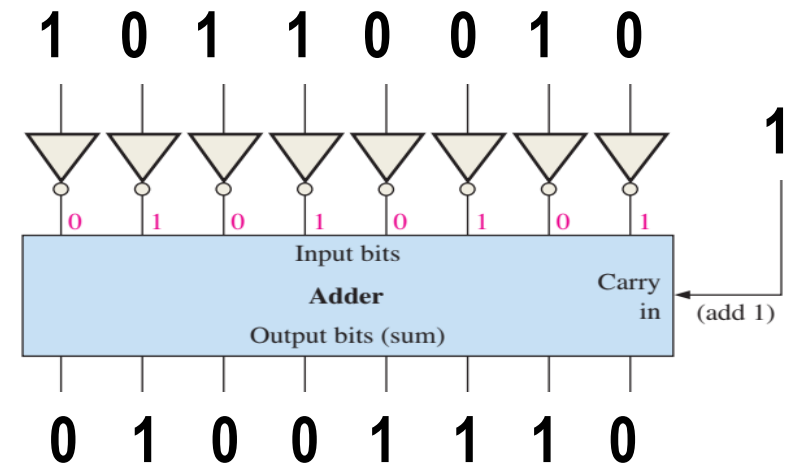
□ 2's complement of a binary number is found by adding 1 to the LSB of the 1's complement

$$\text{2's complement} = (\text{1's complement}) + 1$$

□ Ex.1 . Find the 2's complement of $(10110010)_2$, using the 1st method.

□ In **hardware**, array of inverters can be used with an adder

Binary Number	1	0	1	1	0	0	1	0
	↓	↓	↓	↓	↓	↓	↓	↓
1's complement	0	1	0	0	1	1	0	1
								+
								1
2's complement	0	1	0	0	1	1	1	0



Hardware Digital circuit for 1's complement

Complements of Binary Numbers (3/5)

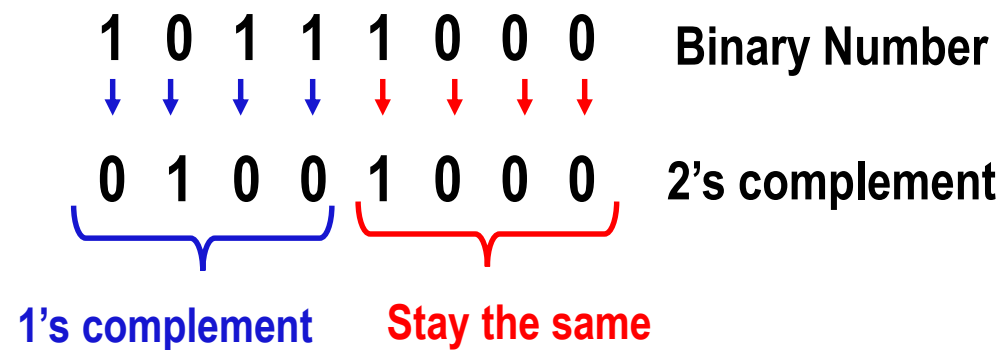


□ 2's Complement

□ 2nd method:-

- Start at the right with the LSB and write the bits as they are up to and including the 1st 1.
- Take the 1's complements of the **remaining bits**.

□ Ex.1 Find the 2's complement of $(10111000)_2$ using the 2nd method



Complements of Binary Numbers (4/5)



❑ The **2's complement** form for representing signed numbers is the most widely used in computers and microprocessor-based systems. **Why?** 

❑ **Using 1st complement**, 0 has **two** different representations.

❑ $(+0)_{10} = (00000000)_2$

❑ $(-0)_{10} = (11111111)_2$

❑ **Using 2nd complement**, 0 has only **one** representation.

❑ $(+0)_{10} = (00000000)_2$

❑ $(-0)_{10} = (00000000)_2$

This is the reason why 2' s complement is generally used.

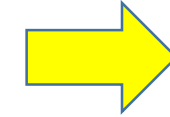


Complements of Binary Numbers (5/5)



❑ **Negative** numbers are the **2's complements** of the corresponding positive numbers.

❑ Ex:- Convert $(-25)_{10}$ into binary using 8-bits.



$(-25)_{10} = (11101111)_2$ 😊

Ans.

Step1: Convert decimal number into binary

Step	quotient	Remainder
25/2	12	1
12/2	6	0
6/2	3	0
3/2	1	1
1/2	0	1

Stop when the Whole-number quotient is 0

1 1 0 0 1

Step2: write the binary number as 8 bits

Step3: find the 1st complements

Step4: find the 2nd complements
= 1st complements+1

	B7	B6	B5	B4	B3	B2	B1	B0
25	0	0	0	1	1	0	0	1
1 st complements	1	1	1	0	0	1	1	0
2 nd complements	1	1	1	0	0	1	1	1



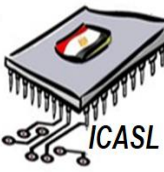
- ☐ Binary Arithmetic
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Hexadecimal Numbers(1/5)

- ❑ The hexadecimal number system has **16** characters {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}
- ❑ **16** characters= 2^4 ← 4bits
- ❑ Each **hexadecimal** digit represents a **4-bit** binary number

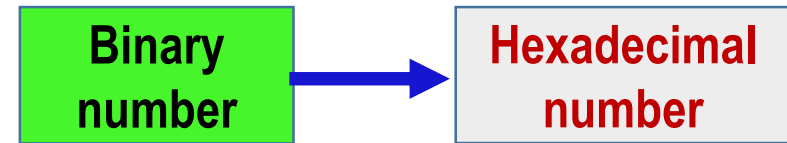
Decimal Number	Binary Number				Hexadecimal
	B3	B2	B1	B0	
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	A
11	1	0	1	1	B
12	1	1	0	0	C
13	1	1	0	1	D
14	1	1	1	0	E
15	1	1	1	1	F



Hexadecimal Numbers(2/5)

❑ Binary-to-Hexadecimal Conversion

- ❑ Break the binary number into 4-bit groups
- ❑ Starting at the right-most bit and replace each 4-bit group with the equivalent hexadecimal symbol.



❑ Ex.1. Convert the following binary number $(1100101001010111)_2$ to hexadecimal

Ans.

$(1100101001010111)_2 \rightarrow (CA57)_{16}$

 C A 5 7

❑ Ex.2. Convert the following binary number $(111111000101101001)_2$ to hexadecimal

Ans.

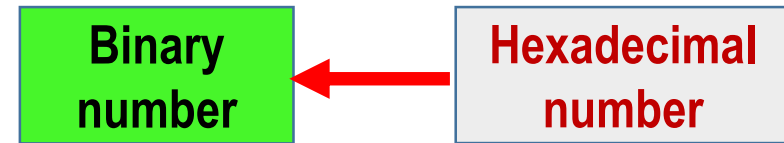
$(00111111000101101001)_2 \rightarrow (3F169)_{16}$

 3 F 1 6 9



Hexadecimal Numbers(3/5)

❑ Hexadecimal-to-Binary Conversion



- ❑ Reverse the process and replace each hexadecimal symbol with the appropriate 4 bits

- ❑ Ex.1. Convert the following hexadecimal number $(10A4)_{16}$ to binary number

Ans.

$$\begin{array}{cccc} 1 & 0 & A & 4 \\ \hline 0001 & 0000 & 1010 & 0100 \end{array} \quad \longrightarrow \quad (0001000010100100)_2$$

- ❑ Ex.2. Convert the following hexadecimal number $(CF8E)_{16}$ to binary number

Ans.

$$\begin{array}{cccc} C & F & 8 & E \\ \hline 1100 & 1111 & 1000 & 1110 \end{array} \quad \longrightarrow \quad (1100111110001110)_2$$

Hexadecimal Numbers(4/5)



□ Hexadecimal-to-Decimal Conversion

- **Step1:** convert the hexadecimal number to binary
- **Step2:** convert the binary to decimal

□ **Ex.1.** Convert the following hexadecimal number $(1C)_{16}$ to decimal number

Ans.

Step1: $\begin{array}{cc} 1 & C \\ \underbrace{} & \underbrace{} \\ 0001 & 1100 \end{array} \longrightarrow = (00011100)_2$

Step2:

Weight	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	↓	↓	↓	↓	↓	↓	↓	↓
Binary Number	0	0	0	1	1	1	0	0
Value in decimal	= 0	+0	+ 0	+1* 2^4	+ 1* 2^3	+1* 2^2	+0	+ 0 = $(28)_{10}$

$$(1C)_{16} = (00011100)_2 = (28)_{10}$$



Hexadecimal Numbers(6/5)



❑ Hexadecimal-to-Decimal Conversion

- ❑ **Step1:** convert the hexadecimal number to binary
- ❑ **Step2:** convert the binary to decimal

❑ **Ex.2.** Convert the following hexadecimal number $(A85)_{16}$ to decimal number

Ans.

Step1: $\begin{matrix} A & 8 & 5 \\ \hline 1010 & 1000 & 0101 \end{matrix} \longrightarrow = (101010000101)_2$

Step2:

Weight	2^{11}	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Binary Number	1	0	1	0	1	0	0	0	0	1	0	1
Value in decimal	$= 1 \cdot 2^{11} + 0 + 1 \cdot 2^9 + 0 + 1 \cdot 2^7 + 1 \cdot 2^2 + 1 \cdot 2^0 = (2693)_{10}$											

$$(A85)_{16} = (101010000101)_2 = (2693)_{10}$$





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Octal Numbers(1/5)

- ❑ The **Octal** number system has **8** digits
{0,1,2,3,4,5,6,7}

❑ $8 = 2^3$ ← 3 bits

- ❑ Each **Octal** digit represents a **3-bit** binary number.

- ❑ It is used **less** frequently than **hexadecimal** 😊
in conjunction with computers and microprocessors

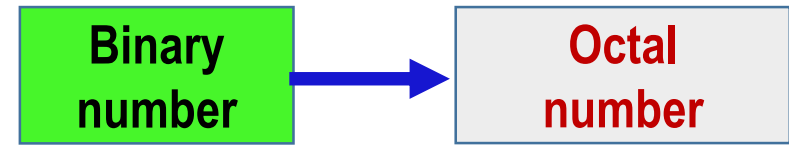
Decimal Number	Binary Number			Octal
	B2	B1	B0	
0	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	1	1	0	6
7	1	1	1	7



Octal Numbers(2/5)

❑ Binary-to-Octal Conversion

- ❑ Break the binary number into 3-bit groups
- ❑ Starting at the right-most bit and replace each 3-bit group with the equivalent octal number



- ❑ Ex.1. Convert the following binary number $(110101)_2$ to octal

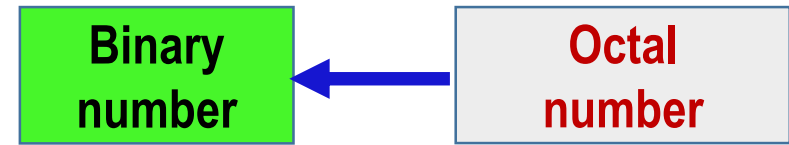
$$\begin{array}{c} (110101)_2 \\ \underbrace{\quad\quad}_6 \underbrace{\quad}_5 \\ \text{Ans.} \quad \longrightarrow \quad (65)_8 \end{array}$$

- ❑ Ex.2. Convert the following binary number $(11010000100)_2$ to octal

$$\begin{array}{c} (011010000100)_2 \\ \underbrace{\quad}_3 \underbrace{\quad}_2 \underbrace{\quad}_0 \underbrace{\quad}_4 \\ \text{Ans.} \quad \longrightarrow \quad (3204)_8 \end{array}$$

Octal Numbers(3/5)

❑ Octal-to-Binary Conversion



❑ Reverse the process and replace each octal with the appropriate 3 bits

❑ Ex.1. Convert the following octal number $(13)_8$ to binary number

Ans.



❑ Ex.2. Convert the following octal number $(140)_8$ to binary number

Ans.



Octal Numbers(4/5)



❑ Octal-to-Decimal Conversion

- ❑ **Step1:** convert the Octal number to binary
- ❑ **Step2:** convert the binary to decimal

❑ **Ex.1.** Convert the following Octal number $(13)_8$ to decimal number

Ans.

Step1:

$$\begin{array}{cc} 1 & 3 \\ \underbrace{} & \underbrace{} \\ 001 & 011 \end{array} \longrightarrow = (001011)_2$$

Step2:

Weight	2^5	2^4	2^3	2^2	2^1	2^0
	↓	↓	↓	↓	↓	↓
Binary Number	0	0	1	0	1	1

$$\text{Value in decimal} = 0 + 0 + 1 \cdot 2^3 + 0 + 1 \cdot 2^1 + 1 \cdot 2^0 = (11)_{10}$$

$$(13)_8 = (001011)_2 = (11)_{10}$$



Octal Numbers(5/5)



❑ Octal-to-Decimal Conversion

- ❑ **Step1:** convert the Octal number to binary
- ❑ **Step2:** convert the binary to decimal

❑ **Ex.2.** Convert the following Octal number $(140)_8$ to decimal number

Ans.

Step1: $\begin{array}{ccc} 1 & 4 & 0 \\ \underbrace{} & \underbrace{} & \underbrace{} \\ 001 & 100 & 000 \end{array} \longrightarrow = (001100000)_2$

Step2:

	Weight	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Binary Number		0	0	1	1	0	0	0	0	0

Value in decimal = $0 + 0 + 1*2^6 + 1*2^5 + 0+0+ 0+0+0 = (96)_{10}$

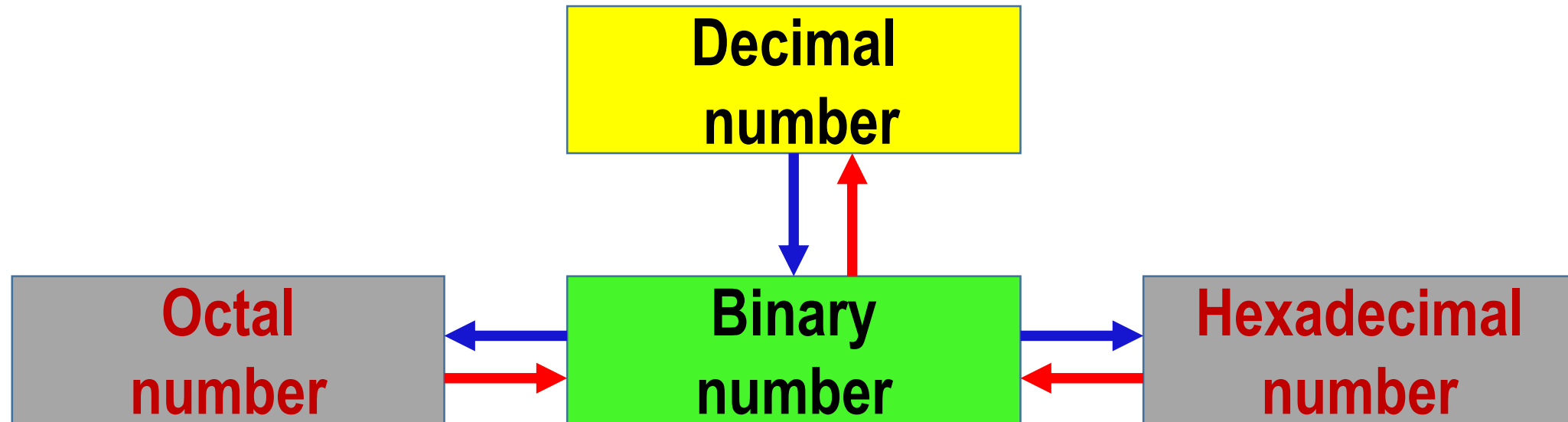
$$(140)_8 = (001100000)_2 = (96)_{10}$$





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Conversion Map



Thanks!